

Einführung in die Kern-, Teilchen und Astrophysik

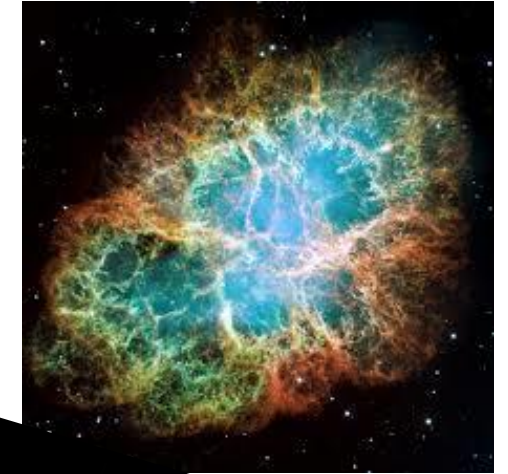
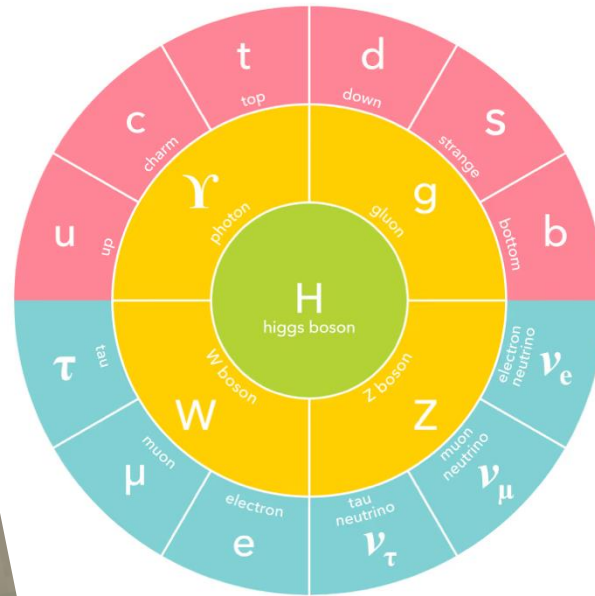
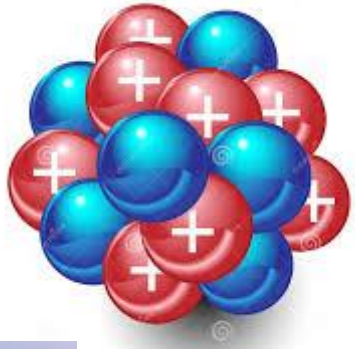


Table of contents

Lecture	Date	Theme	Topic	Sub-Topics
1	14/10/25	Introduction	Little History of particle physics	atoms, elementary particles
2	21/10/25	Particle Physics	Standard Model	Fermions, Bosons, Interactions,
3	28/10/25		Fundamental symmetries	Conservations, Feynman, Propagator,
4	04/11/25		Fundamental symmetries	Discrete symmetries; Cross section
5	11/11/25		Interaction with Matter	Bethe Bloch, gamma interaction
6	18/11/25		Particle detection	Gamma, neutron interactions, detectors
7	25/11/25	Particle/Astrophysics	Accelerators, Cosmic accelerators	Four vector, accelerators, cyclotron Supernovae, cosmic rays
8	02/12/25	Nuclear physics	Basics	Binding energies, Weizäcker
9	09/12/25		Nuclear reactions	Alpha, beta, gamma
10	16/12/25		Nuclear reactors	Radiation, Reactors
11	13/01/26	Particle / Nuclear	High energy physics 1	Deep inelastic scattering
12	20/01/26		High energy physics 2	QCD
13	27/01/26	Particle Physics	Weak interactions	
14	03/02/26	Astrophysics	Neutrinos	



Literatur

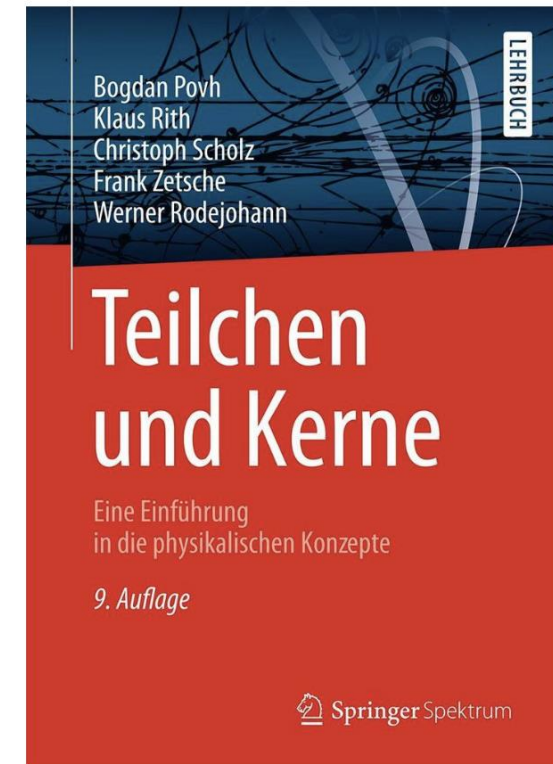
- Terranova, A modern primer in particle & nuclear physics (Oxford Univ. Press)
- Povh, Rith, Scholz, Zetsche; Teilchenphysik (Springer Verlag)
- Perkins; Introduction to High Energy Physics (Addison Wesley)
- Mayer-Kuckuck; Kernphysik (Teubner)
- Kundt; Astrophysics A primer (Springer)
- Hanslmeier: Einführung in Astronomie und Astrophysik (Springer)

Tiefergehende Literatur

- Berger; Elementarteilchenphysik (Springer)
- Halzen, Martin; Quarks & Leptons (Wiley)
- Kleinknecht; Teilchendetektoren (Teubner)
- Carroll, Ostlie; An Introduction to Modern Astrophysics (AddisonWesley)

Literatur:

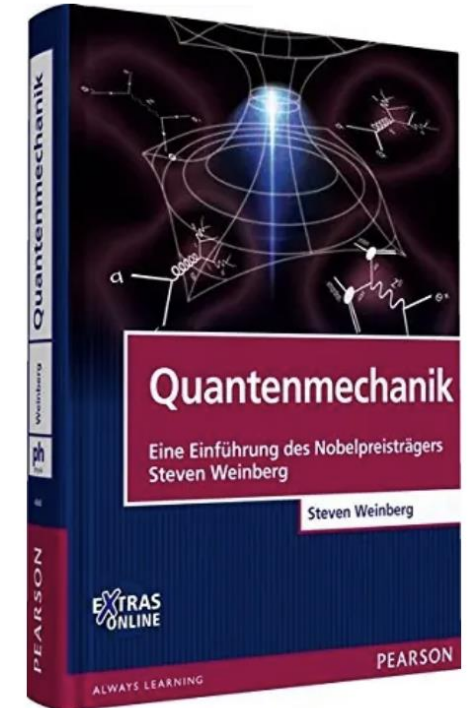
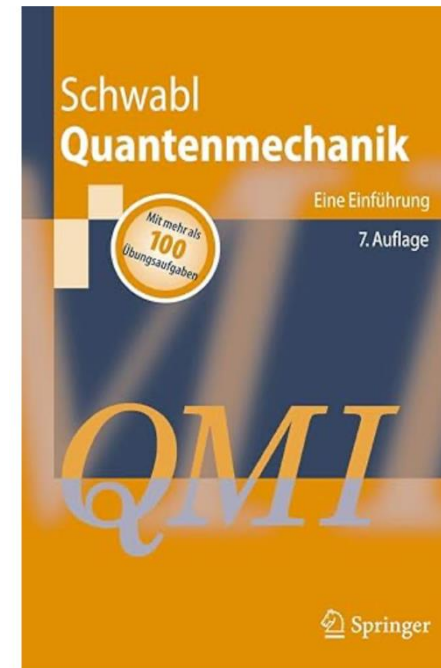
- Die ersten drei Minuten (Steven Weinberg)



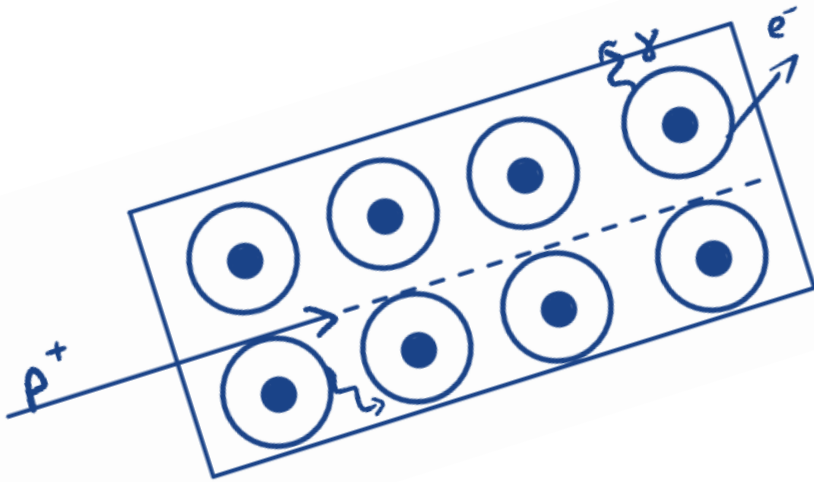
Einführung in Quantenmechanik (für Nicht-Physiker und Physiker 😊)

Einfache und 'unterhaltsame' Einführung in die Quantenphysik:

https://www.didaktik.physik.uni-muenchen.de/archiv/inhalt_materialien/quantenphysik_sek1/quant_sek1.pdf

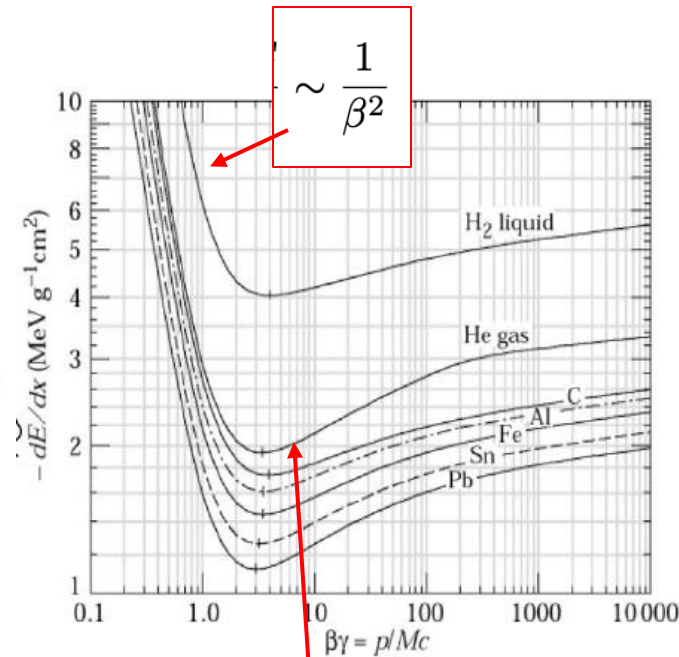


Zusammenfassung der letzten Vorlesung



$$-\frac{1}{\rho} \frac{dE}{dx} = K z^2 \frac{Z}{A} \frac{1}{\beta^2} \left[\ln \left(\frac{2m_e c^2 (\beta\gamma)^2}{I} \right) - \beta^2 \right]$$

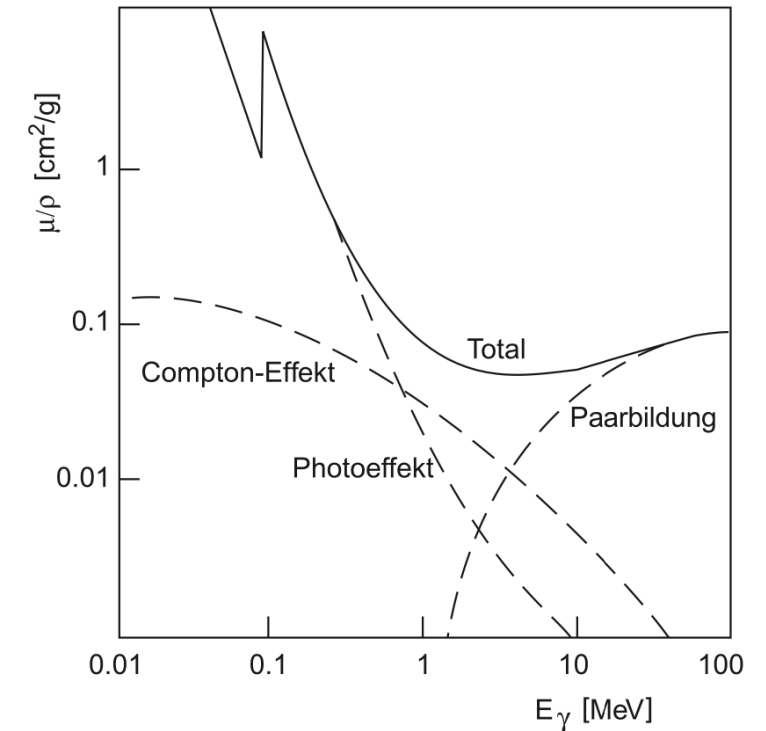
Bethe Bloch



MIP

$$-\frac{1}{\rho} \frac{dE}{dx} \approx 2 \text{ MeV cm}^2/\text{g}$$

Energieverlust in
Materie



Gamma interactions

Nachtrag aus Skript: Energieverlust von Elektronen

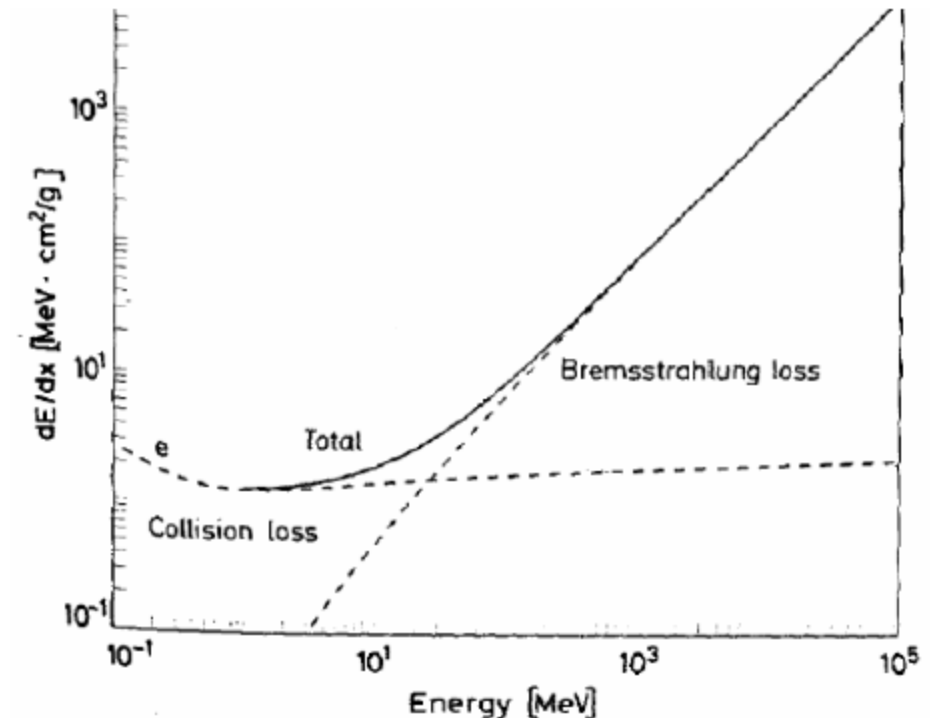
Sonderfall Elektronen/Positronen:

Da e^-/e^+ sehr leicht sind, erfahren sie stärkere Ablenkung im Feld der Kerne. Durch die entsprechende Abbremsung werden Photonen abgestrahlt, die Bremsstrahlung. Diese ist stark energie- und Molekül-abhängig. Zusätzlich zu der oben beschriebenen Bethe-Bloch Formel gibt es einen additiven Faktor zur kompletten Beschreibung von dE/dx :

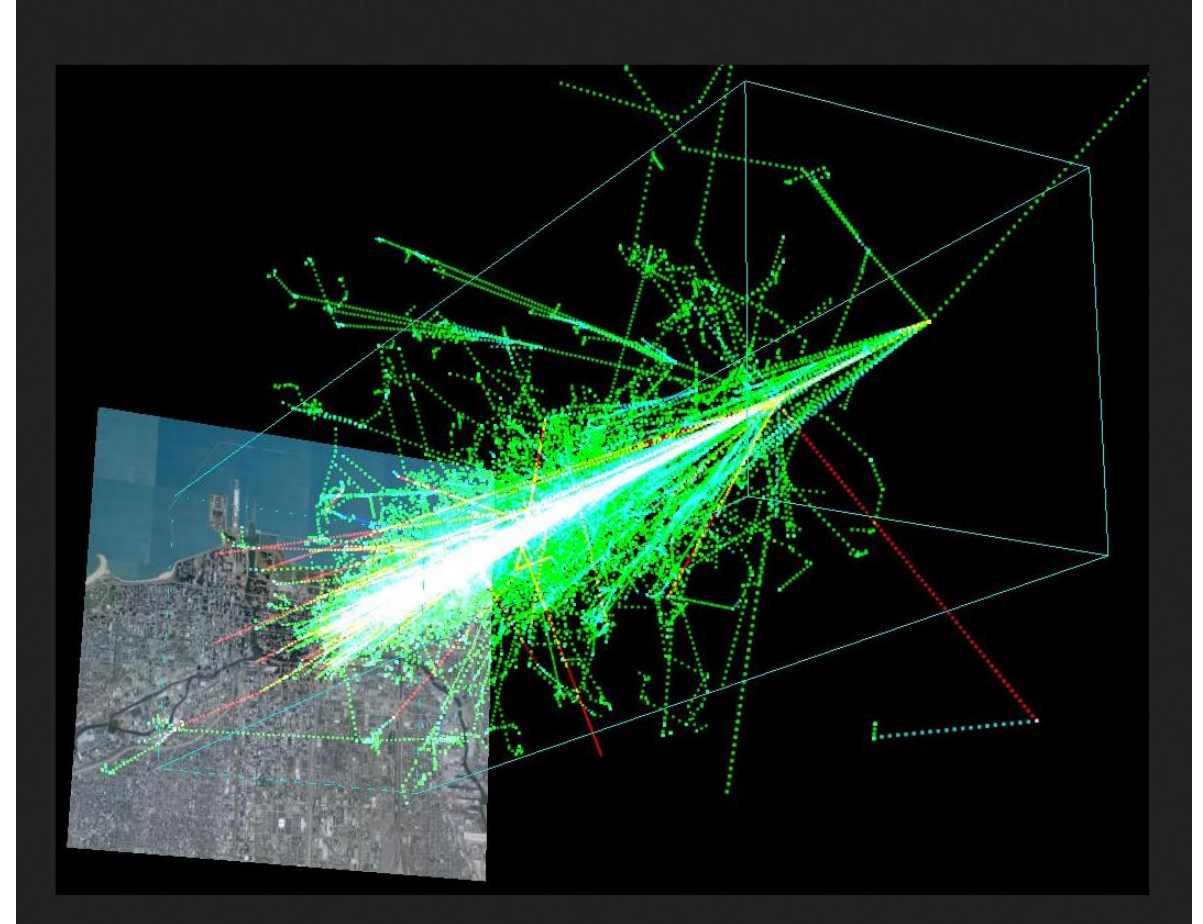
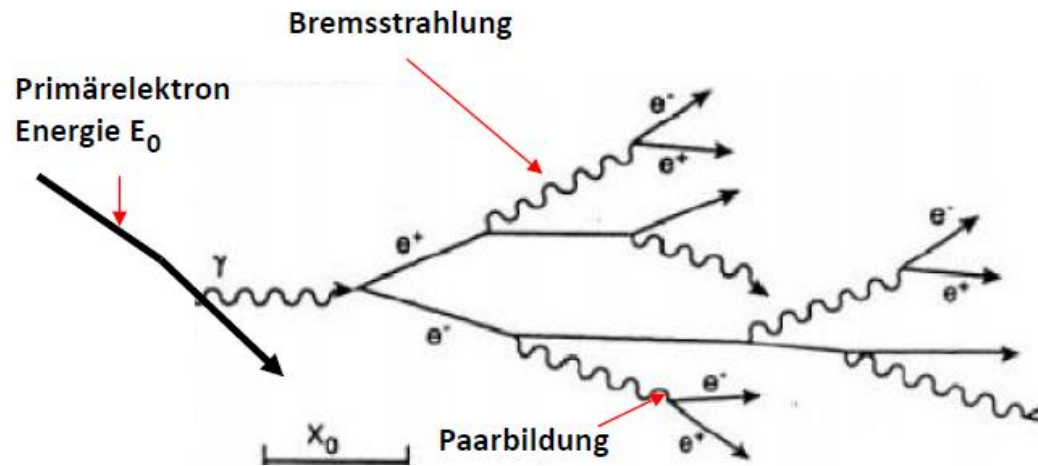
$$\left. \frac{dE}{dx} \right|_{\text{rad}} \sim E \cdot Z^2$$

dominant oberhalb von $E_c \approx 800\text{MeV}/Z$

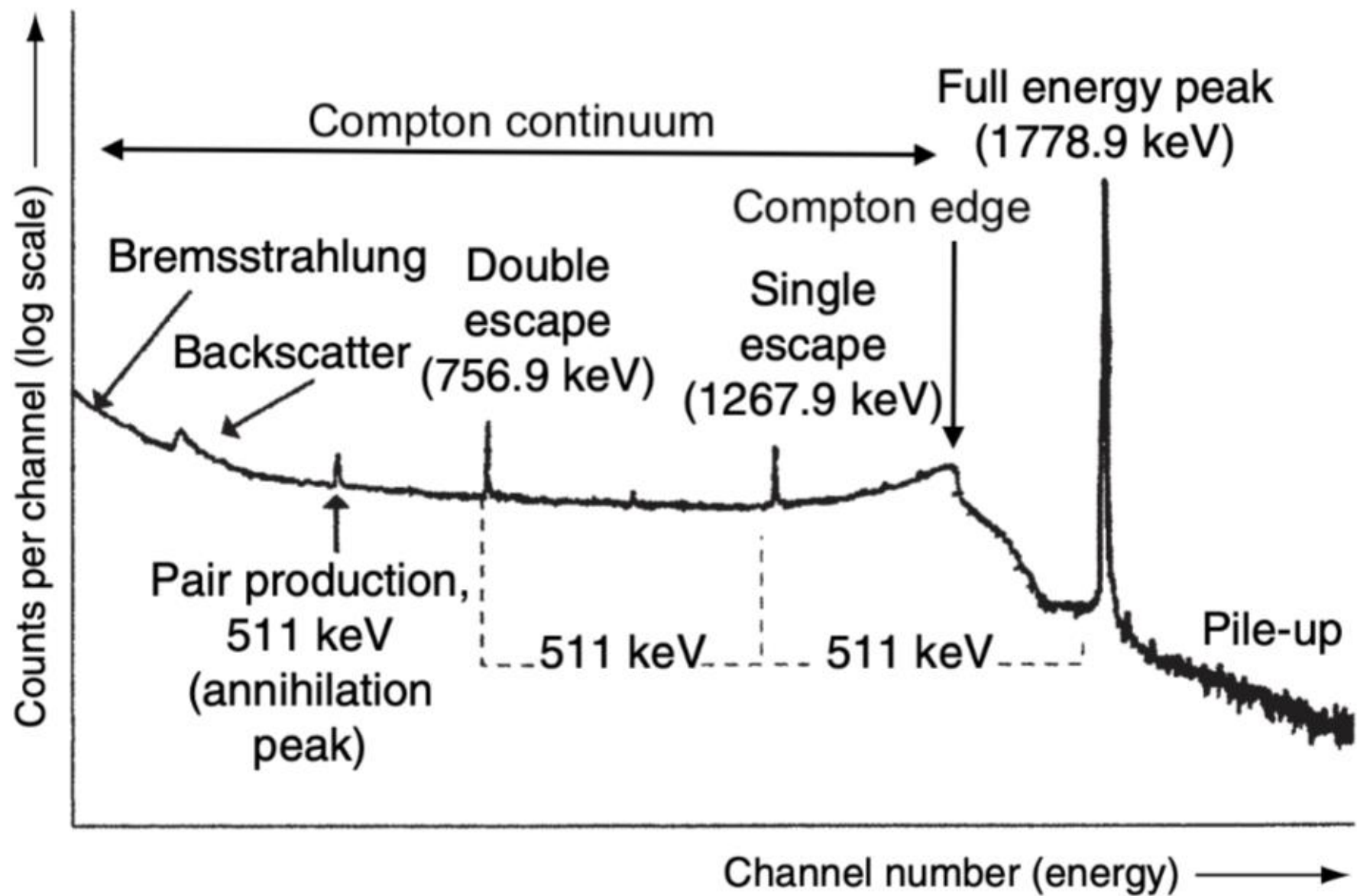
*Energieverlust schneller Elektronen
in Materie*



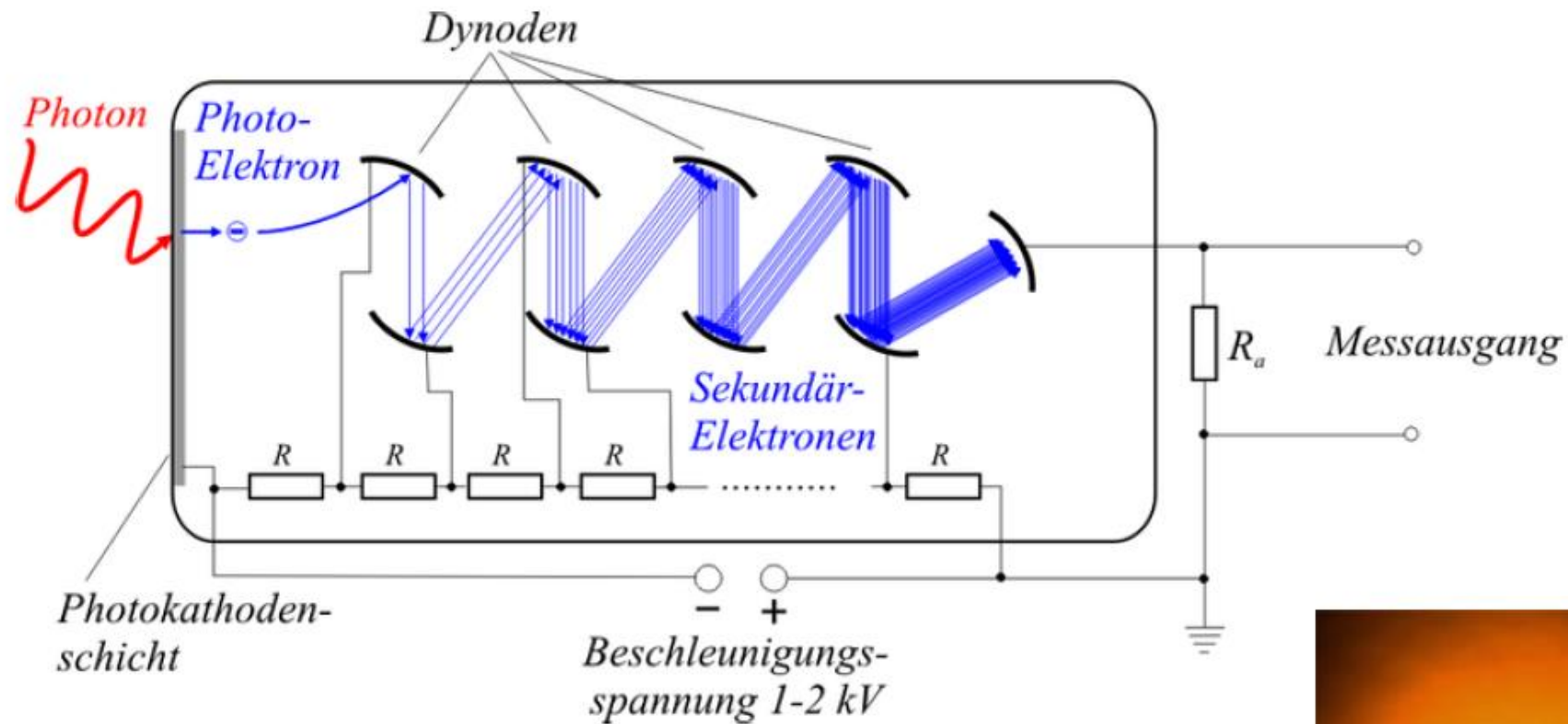
WW hochenergetischer Photonen



Teilchenschauer, verursacht durch kosmische Strahlung in der Atmosphäre.
Hier: 1 TeV Proton, 20km Höhe, Computersimulation



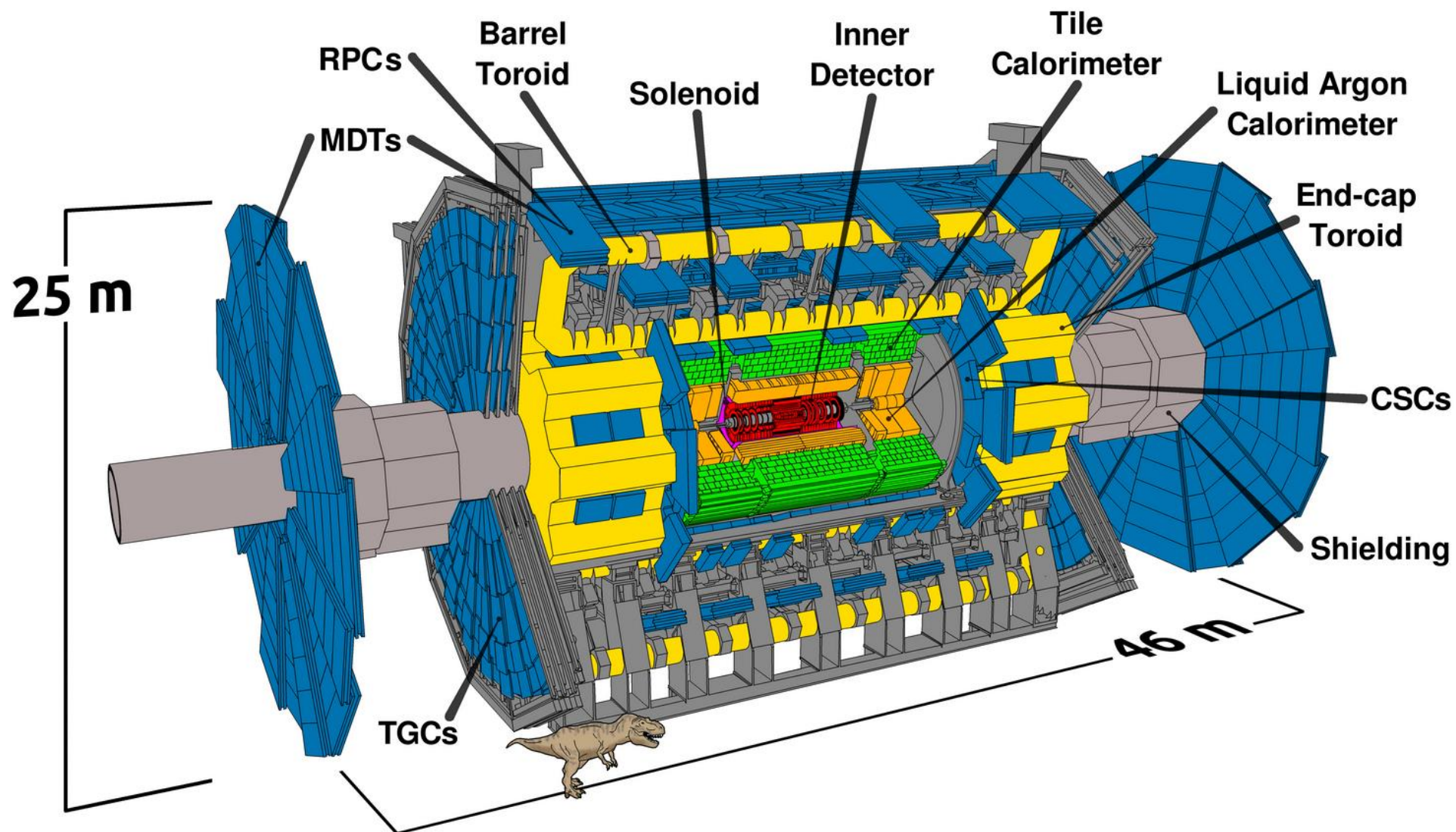
Nachweis von Photonen von Szintillator: Photomultiplier

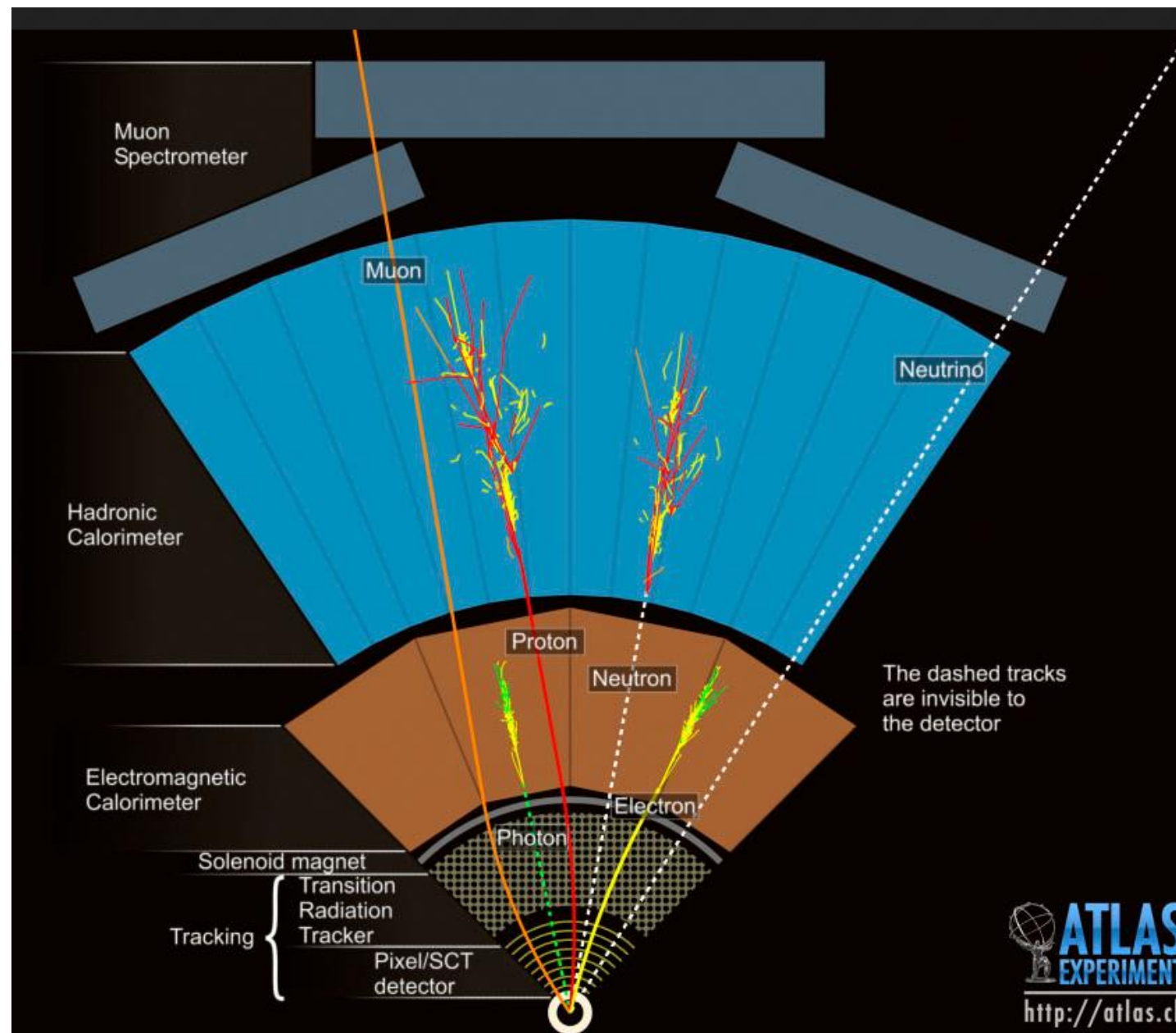


Quantum Efficiency ϵ :
gibt an wieviele Photoelektronen pro Photon erzeugt werden.
Typisch $\epsilon = 0.25$



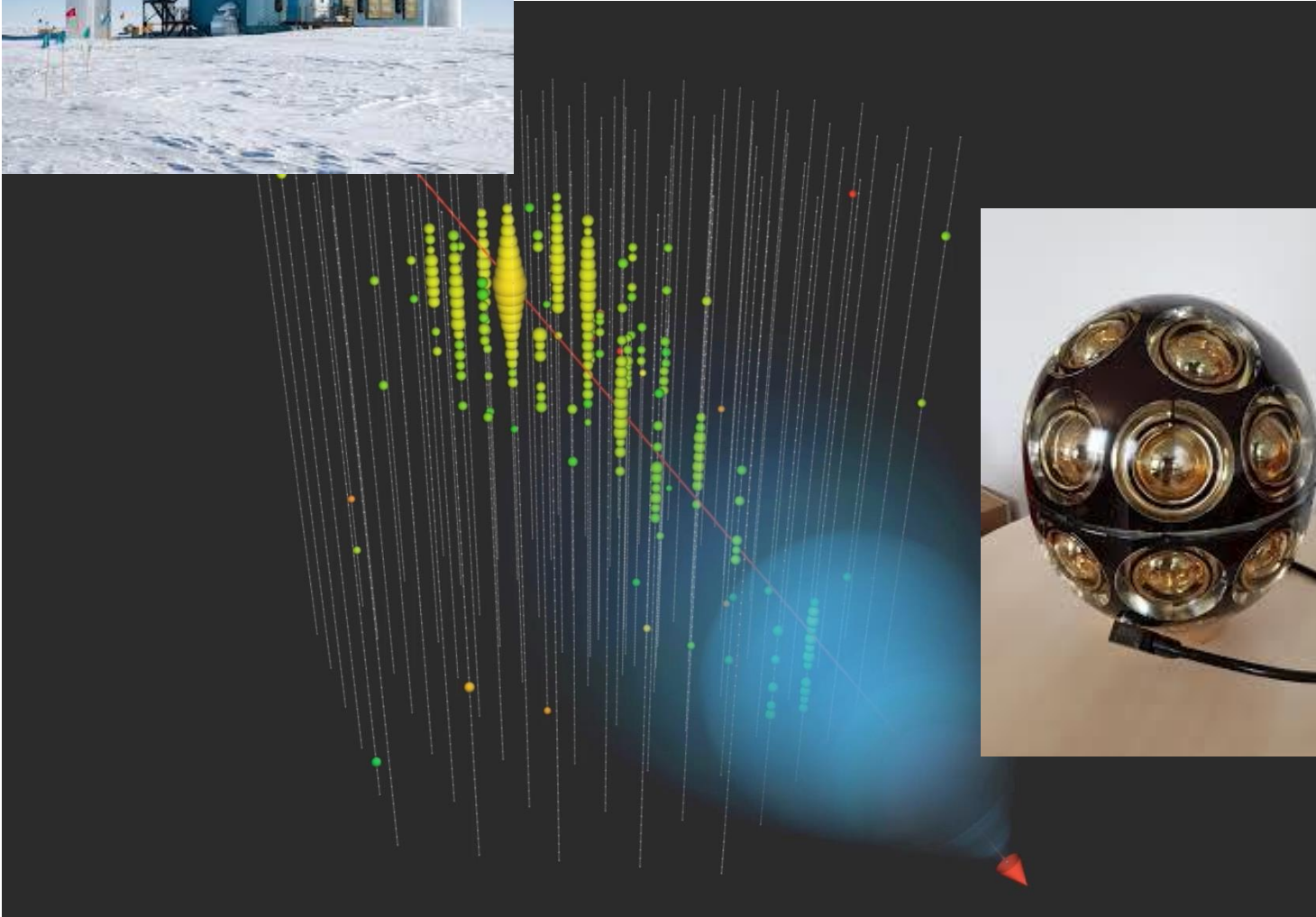
ATLAS Detektor (LHC – CERN)





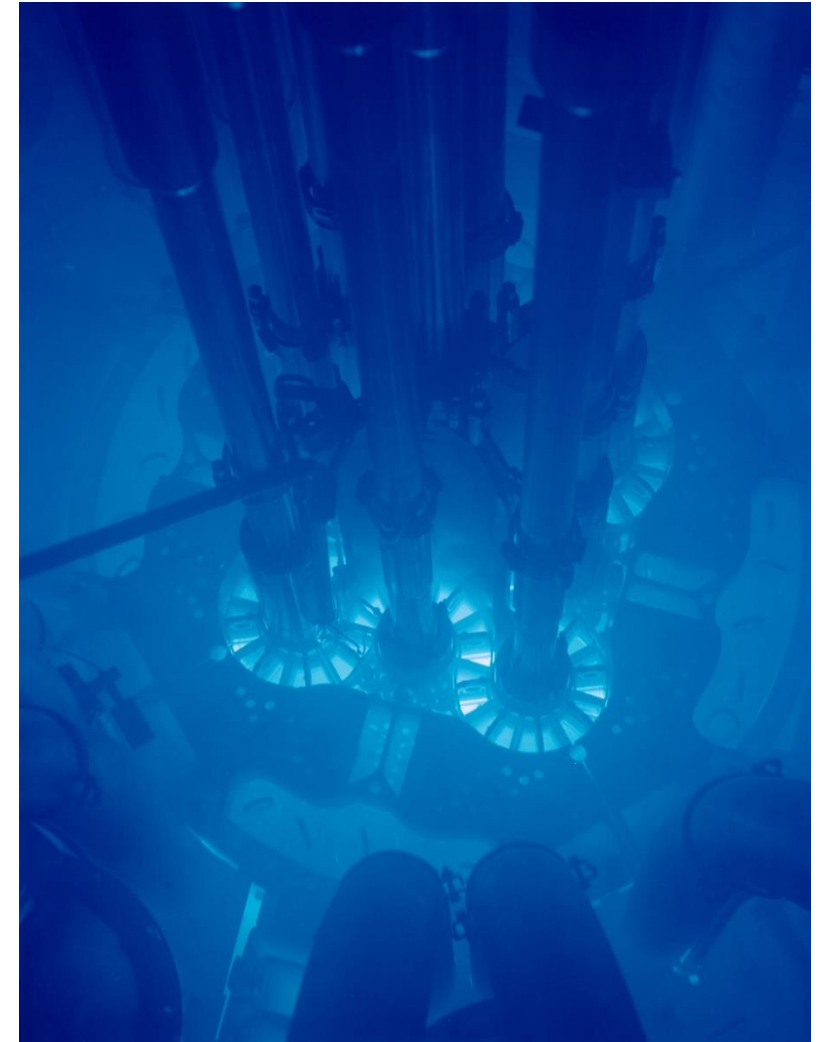


Ice Cube Detector at South Pole



Detect Cherenkov Light in Ice!

Cherenkov light in reactor pool



Cryogenic detectors for neutrino detection

